Required Formula List for the Simulation

A. Feedstock and organic-matter composition

1. Total solids

$$M_{TS} = M_{wet} \times f_{TS}$$

Where:

- M_{TS}= total solids mass
- M_{wet}= wet mass of food waste
- f_{TS}= total solids fraction

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2. Volatile solids

$$M_{VS} = M_{TS} \times f_{VS/TS}$$

Where:

- M_{VS}= volatile solids mass
- $f_{VS/TS}$= volatile solids fraction of total solids

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3. Organic-matter fraction balance

$$f_{carb} + f_{protein} + f_{lipid} + f_{inert} = 1$$

Where:

- f_{carb}= carbohydrate / polysaccharide fraction
- $f_{protein}$= protein / polypeptide fraction
- f_{lipid}= lipid / polylipid fraction
- f_{inert}= non-biodegradable fraction

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4. Mass of each organic-matter class

$$M_{carb} = M_{VS} \times f_{carb}$$
$$M_{protein} = M_{VS} \times f_{protein}$$
$$M_{lipid} = M_{VS} \times f_{lipid}$$
$$M_{inert} = M_{VS} \times f_{inert}$$

B. SCG co-substrate composition

5. SCG and food-waste mass split

$$M_{SCG} = M_{VS,total} \times f_{SCG}$$
$$M_{FW} = M_{VS,total} \times (1 - f_{SCG})$$

Where:

- f_{SCG}= SCG ratio, for example 0, 0.10, 0.20 or 0.30

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6. Mixed-substrate composition

$$f_{i,mix} = (1 - f_{SCG}) f_{i,FW} + f_{SCG} f_{i,SCG}$$

Where:

- i= carbohydrate, protein, lipid or inert
- $f_{i,mix}$= final mixed composition
- $f_{i,FW}$= composition of mixed food waste
- $f_{i,SCG}$= composition of SCG

C. Baseline methane potential

7. Theoretical BMP from organic-matter classes

$$BMP_{mix} = (f_{carb} \times BMP_{carb}) + (f_{protein} \times BMP_{protein}) + (f_{lipid} \times BMP_{lipid})$$

Suggested theoretical BMP values:

$$BMP_{carb} = 0.415$$
$$BMP_{protein} = 0.496$$
$$BMP_{lipid} = 1.014$$

Units:

$$Nm^3 CH_4/kgVS$$

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8. Baseline methane yield with biodegradability

$$Y_{CH_4,control} = BMP_{mix} \times B_{control}$$

Where:

- $Y_{CH_4,control}$= methane yield without fungal pretreatment
- $B_{control}$= baseline biodegradability factor

D. Fungal pretreatment severity

9. Enzyme dose response

$$D_i = \frac{E_i}{K_{E,i} + E_i}$$

Where:

- D_i= enzyme dose-response term
- E_i= enzyme dose or activity for organic-matter class i
- $K_{E,i}$= half-saturation enzyme constant
- i= carbohydrate, protein, lipid or lignocellulosic/phenolic fraction

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10. Temperature response

$$T_i = \exp\left[\quad -\left(\frac{T_{pre} - T_{opt,i}}{\sigma_{T,i}}\right)^2 \right]$$

Where:

- T_{pre}= pretreatment temperature
- $T_{opt,i}$= optimum temperature for enzyme group i
- $\sigma_{T,i}$= temperature tolerance width

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11. pH response

$$P_i = \exp\left[\quad -\left(\frac{pH_{pre} - pH_{opt,i}}{\sigma_{pH,i}}\right)^2 \right]$$

Where:

- pH_{pre}= pretreatment pH

- $pH_{opt,i}$= optimum pH for enzyme group i
- $\sigma_{pH,i}$= pH tolerance width

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12. Time response

$$R_i = 1 - e^{-k_{t,i}t_{pre}}$$

Where:

- R_i= time-response term
- $k_{t,i}$= time-response coefficient
- t_{pre}= pretreatment time

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13. Fungal pretreatment effectiveness for each organic-matter class

$$A_i = D_i \times T_i \times P_i \times R_i$$

Where:

- A_i= fungal enzyme effectiveness for substrate class i

This is useful because it prevents enzyme dose alone from directly predicting methane. The enzyme first changes substrate accessibility and biodegradability.

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14. Overall fungal pretreatment severity index

$$FPSI = w_{carb}A_{carb} + w_{protein}A_{protein} + w_{lipid}A_{lipid} + w_{SCG}A_{SCG}$$

Where:

- $FPSI$= fungal pretreatment severity index
- w_i= weighting factor for each substrate class

E. Pretreatment effect on substrate solubilisation

15. SCOD increase

$$SCOD_{increase}(\%) = \frac{SCOD_{pretreated} - SCOD_{control}}{SCOD_{control}} \times 100$$

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16. Estimated SCOD after pretreatment

$$SCOD_{pretreated} = SCOD_{control} \times (1 + \alpha_{SCOD} FPSI)$$

Where:

- α_{SCOD}= calibration coefficient linking fungal pretreatment to SCOD increase

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17. Soluble sugar release

$$Sugar_{release} = Sugar_{control} \times (1 + \alpha_{sugar} A_{carb})$$

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18. FAN/TAN change

$$FAN_{pretreated} = FAN_{control} \times (1 + \alpha_{FAN} A_{protein})$$
$$TAN_{pretreated} = TAN_{control} \times (1 + \alpha_{TAN} A_{protein})$$

F. Hydrolysis and lag-phase modification

19. Modified hydrolysis rate

$$k_{hyd,modified} = k_{hyd,base}(1 + \alpha_{hyd} FPSI)$$

Where:

- $k_{hyd,modified}$= hydrolysis rate after fungal pretreatment
- $k_{hyd,base}$= baseline hydrolysis rate
- α_{hyd}= calibration coefficient

This is one of the most important formulas for connecting fungal pretreatment to AD simulation.

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20. Lag-phase reduction

$$\lambda_{modified} = \lambda_{base}(1 - \alpha_{\lambda} FPSI)$$

Where:

- $\lambda_{modified}$= lag phase after pretreatment
- λ_{base}= baseline lag phase

- α_λ = lag-reduction coefficient

G. Methane production

21. Biodegradability after fungal pretreatment

$$B_{i,treated} = B_{i,control} + (1 - B_{i,control})A_i$$

Where:

- $B_{i,treated}$ = biodegradability of substrate class i after pretreatment
- $B_{i,control}$ = baseline biodegradability
- A_i = fungal enzyme effectiveness

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22. Methane yield from each substrate class

$$Y_{CH_4,i} = M_i \times BMP_i \times B_i$$

Where:

- M_i = VS mass of substrate class i
- BMP_i = methane potential of substrate class i
- B_i = biodegradability factor

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23. Total methane yield

$$Y_{CH_4,total} = \sum_i Y_{CH_4,i}$$

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24. Methane improvement from fungal pretreatment

$$Improvement(\%) = \frac{Y_{CH_4,treated} - Y_{CH_4,control}}{Y_{CH_4,control}} \times 100$$

H. Methane curve over digestion time

25. Simple first-order cumulative methane curve

$$CH_4(t) = CH_{4,max}\left(1-e^{-k(t-\lambda)}\right)$$

For $t > \lambda$.

Where:

- $CH_4(t)$= cumulative methane at time t
- $CH_{4,max}$= maximum methane potential
- k= methane production rate coefficient
- λ= lag phase

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26. Modified Gompertz model

$$CH_4(t) = CH_{4,max}\exp\left\{-\exp\left[\frac{R_{max}e}{CH_{4,max}}(\lambda - t) + 1\right]\right\}$$

Where:

- R_{max}= maximum methane production rate
- e= Euler's number
- λ= lag phase

This is useful if Mohammad wants a more recognised BMP curve-fitting equation.

I. Biogas and methane purity

27. Methane purity

$$P_{CH_4} = \frac{V_{CH_4}}{V_{biogas}} \times 100$$

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28. Total biogas volume

$$V_{biogas} = \frac{V_{CH_4}}{P_{CH_4}}$$

Use P_{CH_4} as a fraction, for example 0.60, not 60.

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29. CO_2 volume

$$V_{CO_2} = V_{biogas} - V_{CH_4}$$

J. Energy recovery

30. Methane energy recovery

$$E_{CH_4} = V_{CH_4} \times LHV_{CH_4}$$

Where:

$$LHV_{CH_4} = 35.8\ MJ/m^3$$

If methane volume is in litres: - Bita used previously for coffee bean example 10%, 20% and 30%

$$E_{CH_4} = \frac{V_{CH_4}(L)}{1000} \times 35.8$$

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31. Electricity potential

$$E_{electricity} = E_{CH_4} \times \eta_{CHP}$$

Where:

- η_{CHP}= electrical conversion efficiency

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32. Energy improvement

$$Energy\ Improvement(\%) = \frac{E_{treated} - E_{control}}{E_{control}} \times 100$$

K. Stability and inhibition indicators

33. VFA risk

$$R_{VFA} = \frac{VFA}{VFA_{crit}}$$

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34. Ammonia risk

$$R_{NH_3} = \frac{FAN}{FAN_{crit}}$$

or:

$$R_{TAN} = \frac{TAN}{TAN_{crit}}$$

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35. pH risk

$$R_{pH} = \frac{|\ pH_{actual} - pH_{opt}\ |}{pH_{opt}}$$

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36. OLR risk

$$R_{OLR} = \frac{OLR}{OLR_{crit}}$$

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37. Combined stability score

$$Stability = 1 - (w_1 R_{VFA} + w_2 R_{NH_3} + w_3 R_{pH} + w_4 R_{OLR})$$

Where:

- w_1, w_2, w_3, w_4= weighting factors
- stability should be constrained between 0 and 1

L. Optimisation formulas

38. Biomethane optimisation objective

$$Objective = Y_{CH_4} + P_{CH_4} + E_{CH_4} - \lambda R_{total}$$

Where:

- Y_{CH_4}= methane yield
- P_{CH_4}= methane purity
- E_{CH_4}= energy recovery

- R_{total}= total risk score
- λ= penalty weight

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39. Multi-objective score

$$Z = w_1 Y_{CH_4} + w_2 P_{CH_4} + w_3 E_{CH_4} - w_4 Cost - w_5 Risk$$

This is useful for the next scope, where cost and risk/TEA are considered.

M. Cost and risk — next scope

40. Pretreatment cost

$$C_{pre} = C_{enzyme} + C_{time} + C_{mixing} + C_{temperature} + C_{labour}$$

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41. Net energy

$$E_{net} = E_{CH_4} - E_{input}$$

Where:

$$E_{input} = E_{mixing} + E_{heating} + E_{pretreatment}$$

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42. Cost per methane produced

$$C_{CH_4} = \frac{C_{total}}{V_{CH_4}}$$

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43. Net benefit

$$Net\ Benefit = Revenue_{energy} - C_{total}$$

Minimum Formula Set for First Python Version

The simplest working version, use these 10 formulas first:

1. M_{VS}=M_{wet}\times f_{TS}\times f_{VS/TS}

2. f_{carb}+f_{protein}+f_{lipid}+f_{inert}=1
3. BMP_{mix}=(f_{carb}BMP_{carb}) +(f_{protein}BMP_{protein}) +(f_{lipid}BMP_{lipid})
4. A_i=
   \frac{E_i}{K_{E,i}+E_i}
   \times
   \exp\left[-\left(\frac{T_{pre}-T_{opt,i}}{\sigma_{T,i}}\right)^2\right]
   \times
   \exp\left[-\left(\frac{pH_{pre}-pH_{opt,i}}{\sigma_{pH,i}}\right)^2\right]
   \times
   (1-e^{-k_{t,i}t_{pre}})
5. FPSI=\sum_i w_iA_i
6. SCOD_{pretreated}=SCOD_{control}(1+\alpha_{SCOD}FPSI)
7. k_{hyd,modified}=k_{hyd,base}(1+\alpha_{hyd}FPSI)
8. Y_ {CH_4, total} =\sum_i M_iBMP_iB_i
9. Improvement (%) =
   \frac {Y_ {CH_4, treated}-Y_ {CH_4, control}} {Y_ {CH_4, control}} \times100
10. E_{CH_4} =V_{CH_4} \times35.8

feed composition → fungal pretreatment → solubilisation/hydrolysis → methane → energy → risk.